

RAG-Based Customer Support Assistant

Low-Level Design (LLD)

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1. Module-Level Design

The system is divided into the following modules:

1. Document Processing Module

- Loads PDF documents
- Extracts raw text

2. Chunking Module

- Splits text into smaller chunks (e.g., 500–1000 tokens)

3. Embedding Module

- Converts text chunks into vector embeddings using embedding models

4. Vector Storage Module

- Stores embeddings in ChromaDB for fast similarity search

5. Retrieval Module

- Retrieves relevant chunks based on user query

6. Query Processing Module

- Accepts user input and prepares it for processing

7. Graph Execution Module (LangGraph)

- Controls workflow between components

8. HITL Module

- Handles escalation to human agents when required

2. Data Structures

- **Document Representation**

```
{  
  "doc_id": string,  
  "content": text  
}
```

- **Chunk Format**

```
{  
  "chunk_id": string,  
  "text": string,  
  "metadata": { "page": number }  
}
```

- **Embedding Structure**

```
{  
  "chunk_id": string,  
  "embedding": [float values]  
}
```

- **Query-Response Schema**

```
{  
  "query": string,  
  "retrieved_chunks": list,  
  "response": string,  
  "confidence": float  
}
```

- **Graph State Object**

```
{  
  "query": string,
```

```
"context": list,  
"response": string,  
"status": string  
}
```

3. Workflow Design (LangGraph)

The system uses a simple 2-node workflow:

- **Processing Node**
 - Retrieves relevant data
 - Generates response using LLM
- **Output Node**
 - Returns response to user

Edges:

Input → Processing Node → Output Node

State:

The state object carries query, context, and response through nodes.

4. Conditional Routing Logic

The routing layer determines how to handle responses:

- **Answer Generation Criteria**
 - Sufficient relevant chunks found
 - High similarity score
- **Escalation Criteria**
 - Low confidence score
 - No relevant context retrieved

- Complex or ambiguous queries

If conditions are not met, the query is escalated to HITL.

5. HITL Design

- **Trigger Conditions**
 - Low confidence in generated answer
 - Missing or insufficient context
- **Process**
 - Query is forwarded to human agent
 - Human reviews and provides response
- **Integration**
 - Human response is returned to user
 - Can be stored for future learning

6. API / Interface Design

- **Input Format**

```
{  
  "query": "string"  
}
```
- **Output Format**

```
{  
  "response": "string",  
  "source": "LLM/HITL",  
  "confidence": "float"  
}
```
- **Flow**

User → API → Processing → Response

7. Error Handling

- **Missing Data**

- Return fallback response

- **No Relevant Chunks**

- Trigger HITL escalation

- **LLM Failure**

- Retry mechanism or fallback message